

EBN 111 ELECTRICITY AND ELECTRONICS

Book:

Charles K Alexander, Mathew N O Sadiku,
Fundamentals of Electric Circuits,
Sixth Edition, McGraw-Hill, 2017

Web:

<https://www.clickup.up.ac.za/>



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Composition of Marks

• Final Mark

- 50% Written Exam (WE)
- 50% Year/Semester Mark (YM)

• Year Mark (YM)

- 66% Semester Test 1 & 2 (ST)
- 12% Class Tests (CT)
- 12% Practicals (P) (50% required in each practical)
- 10% ClickUp Assignments



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Exam Entrance

- To qualify for the final exam you **MUST** meet all the following conditions:
 - Minimum Year/Semester Mark of 30%
 - Must Write all the scheduled tests
 - Must do all the practical/Lab
- Failure to any of the above will result in **EXAM REFUSAL**
- To pass the exam:
 - Must obtain **MINIMUM of 40%** in written exam, and
 - Must have minimum of 50% Final Mark



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Time Table

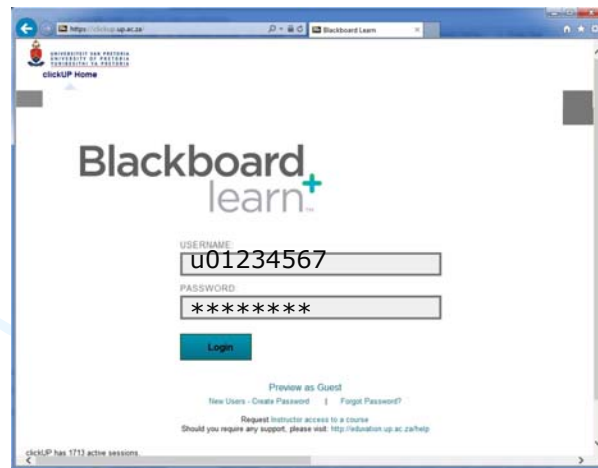
- **Lectures**

	English
– Monday	08:30 Eng 3-1&6
– Tuesday	15:30 Eng 3-1&6
– Thursday	11:30 Eng 3-1&6
– Friday	07:30 Eng 3-1&6 (Tutorial)
- **Practical/Lab**
 - See the web page for details.



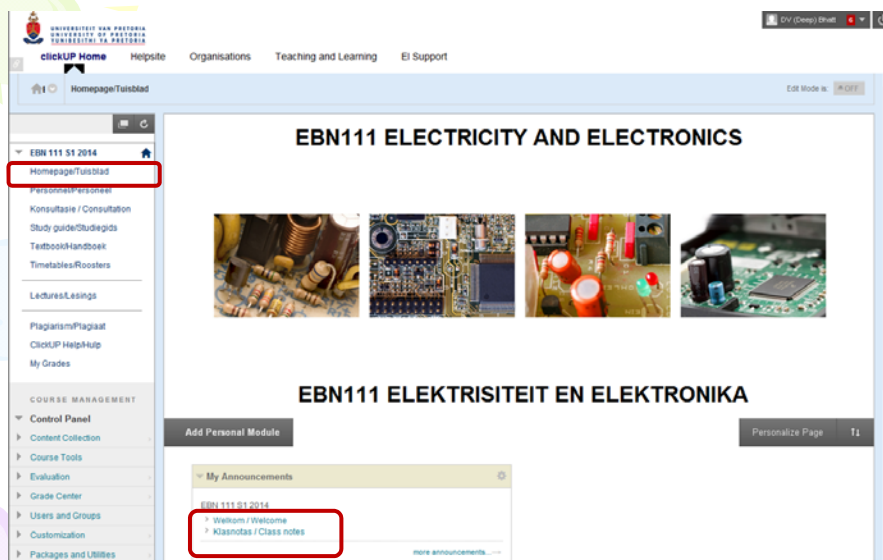
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Announcements

EBN 111 S1 2014

Homepage/Tuisblad
Personnel/Personeel
Konsultasie / Consultation
Study guide/Studiegids
Textbook/Handboek
Timetables/Roosters

Lectures/Lesings
Plagiarism/Plagiaat
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My Grades

COURSE MANAGEMENT

Control Panel

Announcements

Weikom / Welcome

Posted on: Monday, January 20, 2014 1:37:58 PM SAST

Baie welkom in die module EBN111! Die doel van die module is om vir voornemende ingenieurs die basis van elektriese stroombaanteorie te leer. Woon asb die klasse by, vra soveel vrae as moontlik, doen al die huiswerkprobleme, en leer hard vir die toetse en eksamen.

Very welcome to the EBN111 module! The aim of this module is to teach the basics of electric circuit theory to all prospective engineers. Please attend all the classes, ask as many questions as possible, do all your homework problems, and learn hard for the tests and exam.

Klasnotas / Class notes

Posted on: Monday, January 20, 2014 1:39:01 PM SAST

Die klasnotas wat gedurende die lesings gebruik gaan word, is beskikbaar onder die "Lectures/Lesings" skakel. Maak asseblief seker jy laai die notas af en druk dit. Jy gaan dit nodig hê om te volg wat in die klas bespreek word. Tydens julle eerste EBN111 lesing begin ons met Lesing#1 oor Eenhede, Elektriese Lading, Stroom en Spanning.

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Lecture	Chapter #, Lec #	Chapter Sections	Themes
1	Chap 1, Lec#1	1.1 – 1.4	Units, Charge and Current, Voltage
2	Chap 1, Lec#2	1.5 – 1.6	Power and Energy, Circuit Elements
3	Chap 2, Lec#1	2.2 – 2.3	Ohm's Law, Nodes & Branches & Loops
4	Chap 2, Lec#2	2.4	Kirchoff's Law
5	Chap 2, Lec#3	2.5 – 2.6	Series and Voltage division, Parallel and Current division

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EBN 111 ELECTRICITY AND ELECTRONICS

Chapter 1 Basic Concepts

- L#1 { 1.1 Systems of Units.
1.2 Electric Charge.
1.3 Current.
1.4 Voltage.
- L#2 { 1.5 Power and Energy.
1.6 Circuit Elements.



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1.1 System of Units

The derived units commonly used in electric circuit theory

Quantity	Unit	Symbol
Electric charge	Coulomb	C
Current	Ampere	A
Electric potential	Volt	V
Energy, work	Joule	J
Power	Watt	W
Resistance	Ohm	Ω
Conductance	Siemens	S
Inductance	Henry	H
Capacitance	Farad	F
Frequency	Hertz	Hz
Force	Newton	N

Decimal multiples and submultiples of SI units

Factor	Prefix	Symbol
10^9	Giga	G
10^6	Mega	M
10^3	Kilo	k
10^{-3}	Milli	m
10^{-6}	Micro	μ
10^{-9}	Nano	n
10^{-12}	Pico	p



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1.2 Electric Charge

- **Charge** is an electrical property of the atomic particles of which matter consists, measured in **coulombs (C)**.
- The charge $e = 1.602 \times 10^{-19} \text{ C}$
- $1 \text{ C} = 1/e = 6.24 \times 10^{18}$ electrons
- **Law of conservation of Charge:** Charge can neither be created nor destroyed, it can only be transferred.



PP1.1

Calculate the amount of charge represented by six million protons.

$$\begin{aligned}
 Q &= 6 \times 10^6 \times e \\
 &= 6 \times 10^6 \times (1.602 \times 10^{-19} \text{ C}) \\
 &= +9.612 \times 10^{-13} \text{ C} \\
 &= 0.9612 \times 10^{-12} \text{ C} \\
 &= 0.9612 \text{ pC} \\
 &= 961.2 \text{ fC}
 \end{aligned}$$



1.3 Current

- Electric current is the time rate of change of charge, measured in amperes (A)

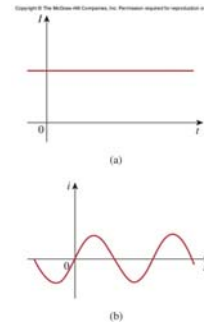
$$i \triangleq \frac{dq}{dt} = \frac{\Delta q}{\Delta t} = \frac{q}{t} \quad \text{-(Can be seen as the slope or gradient)}$$

- Amount of charge transferred between t_0 and t

$$Q \triangleq \int_{t_0}^t i \, dt$$

(Can be seen as the area under the curve)

- charge can vary with time
 - A direct current (dc) is a current that remains constant with time.
 - An alternating current (ac) is a current that varies sinusoidally with time



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P1.6: The charge entering a certain element is shown in Fig. Find the current at:

a) $t=1 \text{ ms}$

b) $t=6 \text{ ms}$

c) $t=10 \text{ ms}$

$$i \triangleq \frac{dq}{dt} = \frac{\Delta q}{\Delta t} = \frac{q}{t}$$

(Can be seen as the slope or gradient)

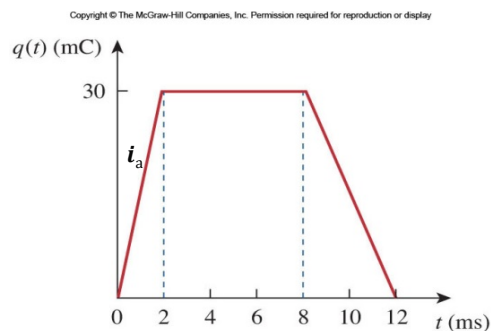
$$i_a \triangleq \frac{30 - 0}{2 - 0} \left(\frac{\text{m}}{\text{ms}} \right) = 15 \text{ A}$$

Or

$$q_a - 30 = \frac{30 - 0}{2 - 0} (t - 2)$$

$$q_a = 15t \quad 0 \leq t \leq 2 \text{ ms}$$

$$i_a = \frac{d}{dt} (15t) = 15 \text{ A}$$



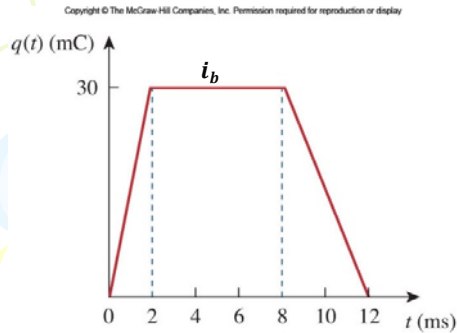
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P1.6: The charge entering a certain element is shown in Fig. Find the current at:

b) $t=6$ ms

c) $t=10$ ms



$$i_b = \frac{30 - 30}{8 - 2} = 0 \text{ A}$$

Or

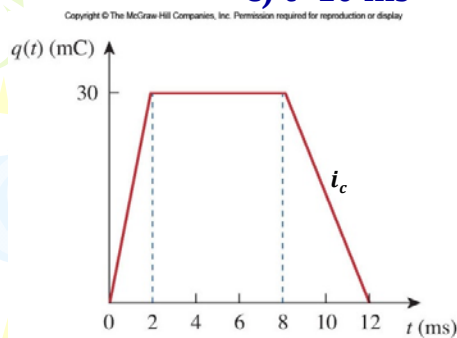
$$q_b - 30 = \frac{30 - 30}{8 - 2}(t - 8)$$

$$q_b = 0 \quad 2 \leq t \leq 8 \text{ ms}$$

$$i_b = \frac{d}{dt}(0) = 0 \text{ A}$$

P1.6: The charge entering a certain element is shown in Fig. Find the current at:

c) $t=10$ ms



$$i_c = \frac{0 - 30}{8 - 2} = -7.5 \text{ A}$$

Or

$$q_c - 0 = \frac{0 - 30}{12 - 8}(t - 12)$$

$$q_c = -7.5t + 90 \quad 8 \leq t \leq 12 \text{ ms}$$

$$i_c = \frac{d}{dt}(-7.5t + 90) = -7.5 \text{ A}$$

$$q = \begin{cases} 15t \text{ C} & 0 \leq t < 2 \text{ s} \\ 30 \text{ C} & 2 \leq t < 8 \text{ s} \\ -7.5t + 90 \text{ C} & 8 \leq t < 12 \text{ s} \end{cases}$$

$$i(t) = \begin{cases} 15 \text{ A} & 0 \leq t < 2 \text{ s} \\ 0 & 2 \leq t < 8 \text{ s} \\ -7.5 \text{ A} & 8 \leq t < 12 \text{ s} \end{cases}$$

**P1.9: The current through an element is shown.
Determine the total charge that passed through
the element at: a) $0 \leq t \leq 1$ s**

$$Q = \int_{t_0}^{t_1} i \, dt$$

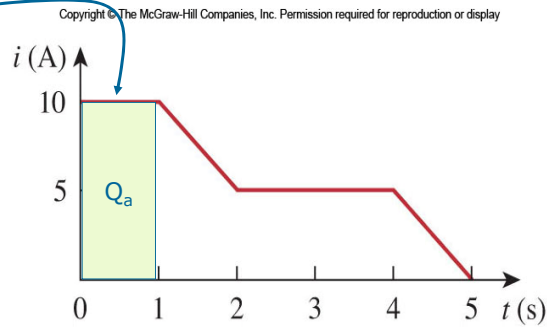
$$i_1 = 10 \, \text{A}$$

$$Q_a = \int_0^1 10 \, dt + q(t_0)$$

$$= [10t]_0^1$$

$$= (10 - 0)$$

$$= \mathbf{10 \, C}$$



Or Q_a is the Area
under curve between
 $0 \leq t \leq 1$ s

$$Q_a = 10 * 1 \\ = \mathbf{10 \, C}$$



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**P1.9: The current through an element is shown.
Determine the total charge that passed through
the element at: b) $0 \leq t \leq 2$ s**

$$Q_b = [q(t_1)=Q_a] + \int_1^2 i_2 \, dt$$

$$Q_a = \mathbf{10 \, C}$$

$$i_2 - 5 = \frac{5 - 10}{2 - 1} (t - 2)$$

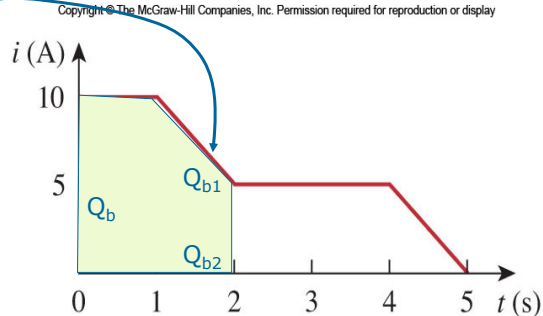
$$i_2 = 15 - 5t$$

$$Q_b = \mathbf{10} + \int_1^2 (15 - 5t) \, dt$$

$$= 10 + [15t - 2.5t^2]_1^2$$

$$= 10 + (30 - 10) - (15 - 2.5)$$

$$= \mathbf{17.5 \, C}$$



Or Q_b is the Area
under curve between
 $0 \leq t \leq 2$ s

$$Q_b = Q_a + Q_{b1} + Q_{b2} \\ = 10 + \frac{1}{2} 5 * 1 + 5 * 1 \\ = \mathbf{17.5 \, C}$$



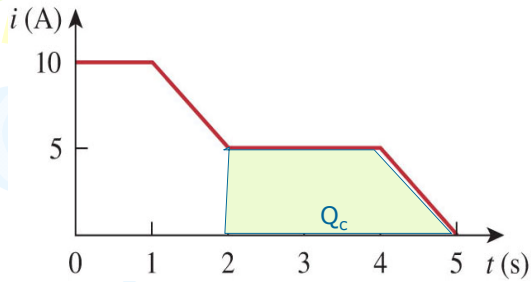
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**P1.9: The current through an element is shown.
Determine the total charge that passed through
the element at:**

c) $2 \leq t \leq 5 \text{ s}$

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Q_c : Area under curve
between $2 \leq t \leq 5 \text{ s}$

$$\begin{aligned} Q_c &= 5 * 2 + \frac{1}{2} 5 * 1 \\ &= 10 + 2.5 \\ &= \mathbf{12.5 \text{ C}} \end{aligned}$$

**PP1-2: The total charge entering a
terminal is given by $q = (10 - 10e^{-2t}) \text{ mC}$.
Find the current at $t = 1 \text{ s}$.**

$$\begin{aligned} i &= \frac{dq}{dt} \\ &= \frac{d}{dt} (10 - 10e^{-2t}) \\ &= -10 * -2 * e^{-2t} \Big|_{t=1} \\ &= 20e^{-2} \\ &= \mathbf{2.7 \text{ A}} \end{aligned}$$

P1.12: If the current flowing through an element is given by

$$i(t) = \begin{cases} 3t \text{ A} & 0 \leq t < 6 \text{ s} \\ 18 \text{ A} & 6 \leq t < 10 \text{ s} \\ -12 \text{ A} & 10 \leq t < 15 \text{ s} \\ 0 & 15 \leq t \end{cases}$$

Plot the charge stored in the element over $0 < t < 20$

$$Q = \int_{t_0}^{t_1} i \, dt + q(0)$$

$$Q_1 = \int_0^6 3t \, dt + 0$$

$$= \left[3 \frac{t^2}{2} \right]_0^6$$

$$= 1.5t^2$$

$$Q_1 = [1.5(36 - 0)]$$

$$Q_1(6) = 54 \text{ C}$$

$$Q_2 = \int_6^t 18 \, dt + 54$$

$$= [18t]_6^t + 54$$

$$= [18t - 108] + 54$$

$$Q_2 = 18t - 54$$

$$Q_2(10) = 126 \text{ C}$$

$$Q_3 = \int_{10}^t -12 \, dt + 126$$

$$= [-12t]_{10}^t + 126$$

$$Q_3 = -12t + 246$$

$$Q_3(15) = 66 \text{ C}$$

$$Q_4 = \int_{15}^t 0 \, dt + 66$$

$$Q_4(20) = 66 \text{ C}$$



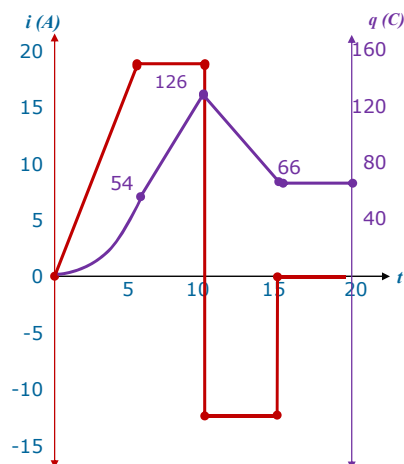
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P1.12: Plot the charge stored in the element over $0 < t < 20$

$$i(t) = \begin{cases} 3t \text{ A} & 0 \leq t < 6 \text{ s} \\ 18 \text{ A} & 6 \leq t < 10 \text{ s} \\ -12 \text{ A} & 10 \leq t < 15 \text{ s} \\ 0 & 15 \leq t \end{cases}$$

$$Q = \begin{cases} 1.5t^2 \text{ C} & 0 \leq t < 6 \text{ s} \\ 18t - 54 \text{ C} & 6 \leq t < 10 \text{ s} \\ -12t + 246 \text{ C} & 10 \leq t < 15 \text{ s} \\ 66 \text{ C} & 15 \leq t \end{cases}$$



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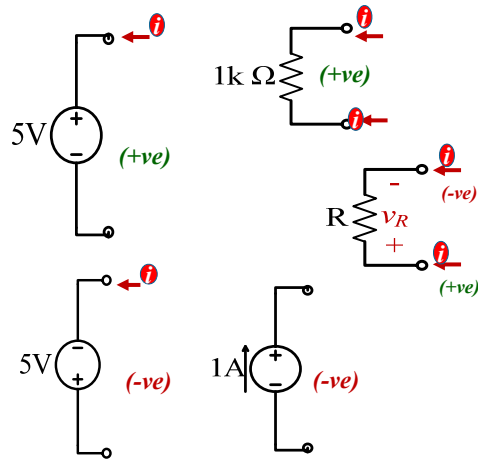
Passive Sign Convention

Current is +ve

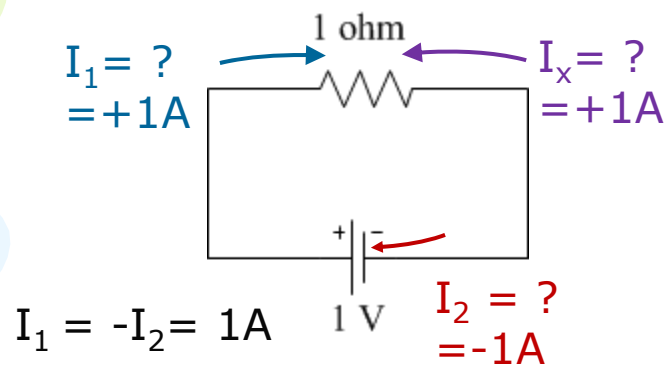
- when current ENTERs through +ve and EXITs through -ve terminal/node of the element

Current is -ve

- When current ENTERs through -ve and EXITs through +ve terminal/node of the element



1.3 Current (3)



1.4 Voltage

- Voltage (or potential difference) is the **energy (work)** required to move a **unit charge** through an element, measured in volts (V).
- Work is done by the electromotive force (*emf*) aka **voltage** or **potential difference**.
- Mathematically:
 - w is energy in joules (J) and q is charge in coulomb (C).
 - voltage is change in energy per charge

$$v_{ab} = \frac{dw}{dq} = \frac{\Delta w}{\Delta q} = \frac{w}{q}$$

$$v_{ab} = v_a - v_b$$



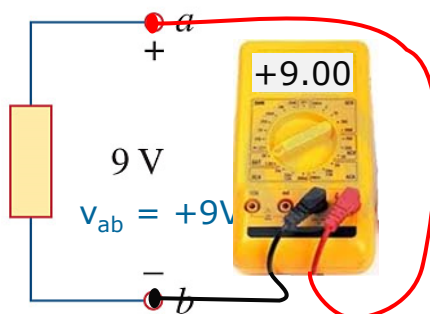
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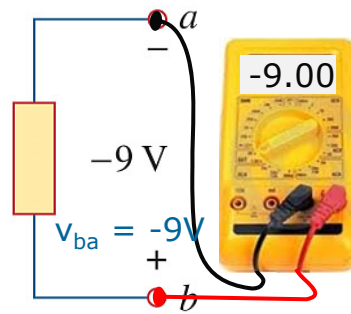
Voltage

Electric voltage, v_{ab} , is always **across the circuit element** or **between two points in a circuit**.

- $v_{ab} > 0$ means the potential of **a** is higher than potential of **b**.
- $v_{ab} < 0$ means the potential of **a** is lower than potential of **b**.
- $v_{ab} = -v_{ba}$



(a)



(b)



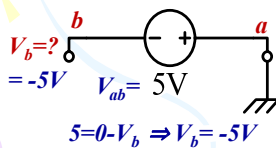
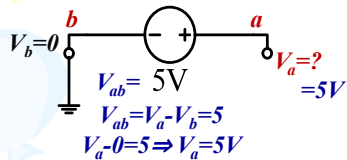
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Voltage, Potential Difference, or *emf*

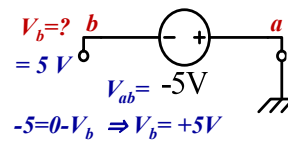
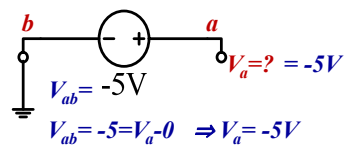
$$\rightarrow V_{ab} = V_a - V_b$$

- A **+ve** value indicates that potential at V_a is HIGHER than V_b



$$\rightarrow V_{ab} = V_a - V_b$$

- A **-ve** value indicates that potential at V_a is LOWER than V_b



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Lecture #2 1.5 Power

- **Power** is the **rate** at which **energy** is consumed or delivered: Joules per second = Watt; [J/s = W].

• Mathematical expression:
$$p = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = v \cdot i$$

- DC systems: $p(t) = P = VI = \text{constant}$

- AC systems: $p(t) = v(t)i(t) = \text{time function}$

The law of **conservation of energy** implies that in an electrical circuit $P_{\text{delivered}} = P_{\text{absorbed}}$ or:

$$\sum p = 0$$



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1.5 Energy

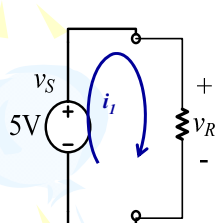
- **Energy** is the capacity to do work, measured in joules (J) or alternatively in Watt-hour (Wh).
- 1 Wh = 3600 J = 3.6 kJ
- 1 kWh = 3.6 MJ
- Energy absorbed or delivered by an element between times t_0 and t_1 is: $w = \int_{t_0}^t p \, dt = \int_{t_0}^t v \cdot i \, dt$
- An element consumes energy at a rate of 1 W for 1 hour – how much energy is that in Joule?
- $w = \int_0^{3600} p \, dt = \int_0^{3600} 1 \, dt = 3600 \text{ J}$



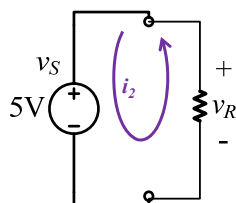
Effect of changing polarity and direction

Calculate v_R ; i_1 ; P_{i1} ; i_2 ; P_{i2} , $R=5 \, \Omega$

Case 1: clockwise or counter-clockwise loop



$$\begin{aligned}
 -v_S + v_R &= 0 \\
 -5 + v_R &= 0 \\
 v_R &= 5 \text{ V} \\
 i_1 &= +\left(\frac{v_R}{R}\right) \\
 &= \frac{5}{5} = 1 \text{ A} \\
 P_{R,i1} &= +i_1 * v_R \\
 &= +1 * 5 \\
 &= +5 \text{ W} \\
 P_{5V,i1} &= -i_1 * v_S \\
 &= -1 * 5 \\
 &= -5 \text{ W}
 \end{aligned}$$



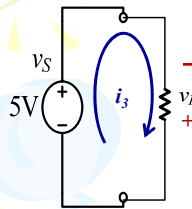
$$\begin{aligned}
 -v_R + v_S &= 0 \\
 -v_R + 5 &= 0 \\
 v_R &= 5 \text{ V} \\
 i_2 &= -\left(\frac{v_R}{R}\right) \\
 &= -\frac{5}{5} = -1 \text{ A} \\
 P_{R,i2} &= -i_2 * v_R \\
 &= -(-1) * 5 \\
 &= +5 \text{ W} \\
 P_{5V,i2} &= i_2 * v_S \\
 &= (-1) * 5 \\
 &= -5 \text{ W}
 \end{aligned}$$



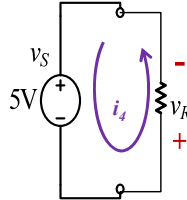
Effect of changing polarity and direction

Calculate v_R ; i_1 ; P_{i1} ; i_2 ; P_{i2} , $R=5\ \Omega$

Case 2: changing the polarity of resistor voltage, v_R



$$\begin{aligned} -v_S - v_R &= 0, \\ -5 - v_R &= 0 \\ v_R &= -5\text{ V} \\ i_3 &= -\left(\frac{v_R}{R}\right) \\ &= -\frac{-5}{5} = 1\text{ A} \\ P_{R,i3} &= -i_3 * v_R \\ &= -1 * -5 \\ &= +5\text{ W} \\ P_{5V,i3} &= -i_3 * v_S \\ &= -1 * 5 \\ &= -5\text{ W} \end{aligned}$$



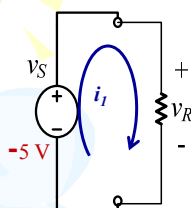
$$\begin{aligned} v_R + v_S &= 0, \\ v_R + 5 &= 0 \\ v_R &= -5\text{ V} \\ i_4 &= \left(\frac{v_R}{R}\right) \\ &= \frac{-5}{5} = -1\text{ A} \\ P_{R,i4} &= +i_4 * v_R \\ &= +(-1) * -5 \\ &= +5\text{ W} \\ P_{5V,i4} &= +i_4 * v_S \\ &= +(-1) * 5 \\ &= -5\text{ W} \end{aligned}$$



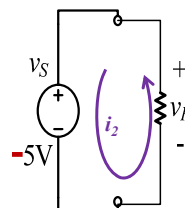
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..calculate v_R ; i_1 ; P_{i1} ; i_2 ; P_{i2} , $R=5\ \Omega$:

Case 3: changing the sign of source



$$\begin{aligned} -v_S + v_R &= 0 \\ -(-5) + v_R &= 0 \\ v_R &= -5\text{ V} \\ i_1 &= +\left(\frac{v_R}{R}\right) \\ &= \frac{-5}{5} = -1\text{ A} \\ P_{i1} &= +i_1 * v_R \\ &= +(-1) * -5 \\ &= +5\text{ W} \end{aligned}$$



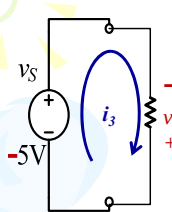
$$\begin{aligned} -v_R + v_S &= 0 \\ -v_R + (-5) &= 0 \\ v_R &= -5\text{ V} \\ i_2 &= -\left(\frac{v_R}{R}\right) \\ &= -\frac{-5}{5} \\ &= +1\text{ A} \\ P_{i2} &= -i_2 * v_R \\ &= -(1) * -5 \\ &= +5\text{ W} \end{aligned}$$



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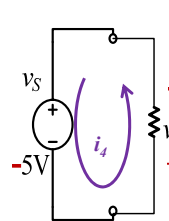
..calculate v_R ; i_1 ; P_{i1} ; i_2 ; P_{i2} , $R=5\ \Omega$:

Case 4: changing the polarity of resistor voltage, v_R



$$\begin{aligned}
 -v_S - v_R &= 0 \\
 -(-5) - v_R &= 0 \\
 v_R &= 5\text{ V} \\
 i_3 &= -\left(\frac{v_R}{R}\right) \\
 &= -\frac{5}{5} \\
 &= -1\text{ A}
 \end{aligned}$$

$$\begin{aligned}
 P_{i1} &= -i_3 * v_R \\
 &= -(-1) * 5 \\
 &= +5\text{ W}
 \end{aligned}$$



$$\begin{aligned}
 v_R + v_S &= 0 \\
 v_R + (-5) &= 0 \\
 v_R &= 5\text{ V} \\
 i_4 &= \left(\frac{v_R}{R}\right) \\
 &= \frac{5}{5} = 1\text{ A}
 \end{aligned}$$

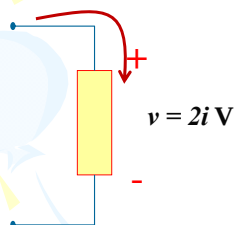
$$\begin{aligned}
 P_{i4} &= +i_4 * v_R \\
 &= +(-1) * 5 \\
 &= +5\text{ W}
 \end{aligned}$$



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PP1-5: Find the power delivered to an element at $t=5\text{ms}$ if the voltage $v = 2i\text{ V}$, the current entering its positive terminal is: $i = 5 \cos 60\pi\text{ A}$.

$$i = 5 \cos 60\pi\text{ A}$$



$$\begin{aligned}
 p &= +i v \\
 &= (i)(2i) \\
 &= 2 * (5 \cos 60\pi)^2 \\
 &= 50 \cos^2(60 * \pi * 5 * 10^{-3}) \\
 &= 17.27\text{ W} \quad \text{radians} \\
 \therefore p > 0: \\
 \therefore \text{Absorbs power}
 \end{aligned}$$

NB: Calculator must be set to radians!
 $2\pi[\text{rad}] = 360^\circ$

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PP1-6

A stove element draws 15 A when connected to a 240 V supply. How long does it take to consume 180 kJ?

$$p = \Delta w / \Delta t = w/t$$

$$\therefore t = w/p = w / (v i)$$

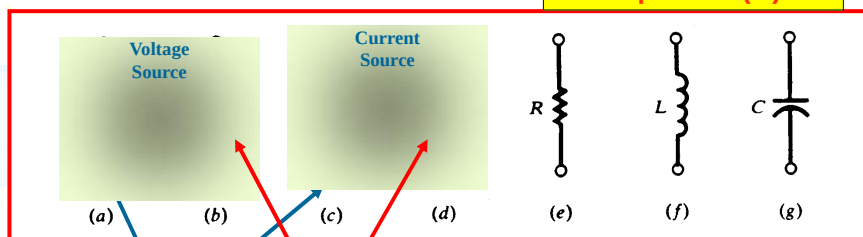
$$t = 180 \times 10^3 / (240 \times 15) \\ = 50 \text{ s}$$



1.6 Circuit Elements (1)

Active Elements (Energy Sources)
Generators, batteries, op-amps

Passive Elements
Resistor (R),
Inductor (L),
Capacitor (C)



Independent
sources

Dependant
sources

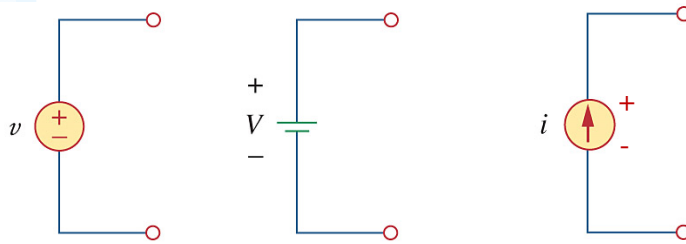
- A dependent source is an active element in which the source quantity is controlled by another voltage or current.
- They have four different types: VCVS, CCVS, VCCS, CCCS. Keep in mind the signs of dependent sources.



Ideal Independent Sources:

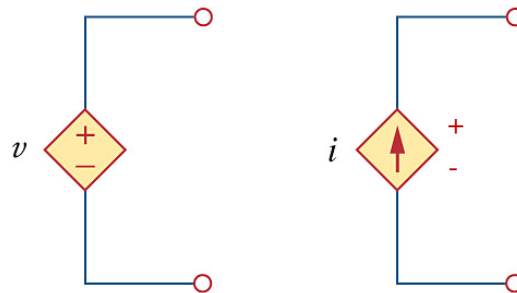
An active element that provides a specified voltage or current that is completely independent of other circuit elements. **ROUND** in shape.

i.e. If $v = 10\text{ V}$ it stays 10 V no matter what!
If $i = 5\text{ A}$ it stays 5 A no matter what



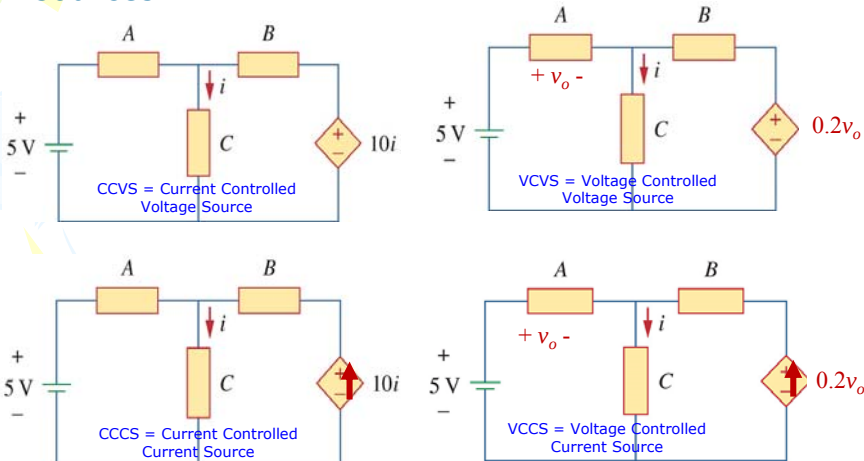
Ideal Dependent Sources:

- An active element in which the source quantity is controlled by another voltage or current. **DIAMOND** in shape.
- Useful in modeling elements such as transistors, operational amplifiers and integrated circuits.



Types of Dependent Sources:

Four possible types of dependent (or controlled) sources:



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Example 1.7: Calculate the power supplied or absorbed by each element.

$$P = \pm i v \text{ (W)}$$

$$P_1 = ?$$

$$P_1 = -i \cdot 5$$

$$= -8 \cdot 5$$

$$= -40 \text{ W}$$

delivered

$$P_2 = +8 \cdot 2 = +16 \text{ W}$$

consumed

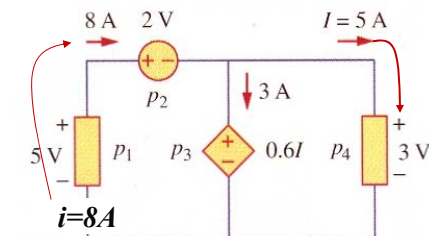


Figure 1.16

$$P_3 = +3 \cdot (0.6I)$$

$$= +3 \cdot (0.6 \cdot 5) = +9 \text{ W}$$

consumed

Power is:

$+ve$ or >0 : absorbed/consumed

$-ve$ or <0 : supplied/delivered

$$P_4$$

$$= +5 \cdot 3$$

$$= +15 \text{ W}$$

consumed

$$\sum_{n=1}^4 P_n = 0$$

$$-40 + 16 + 9 + 15 = ?$$



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